

INDIAN OIL CORPORATION LIMITED // REFINERIES DIVISION

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NOTE FOR APPROVAL (NFA)

File No: IOCL/GHR/HCU-11/NFA/2026-09/042 | Date: 15-September-2026 | Unit: Hydrocracker (HCU-11)
Initiating Department: Inspection & Asset Integrity Division

SUBJECT: Emergency Repair Sanction — Statutory ASME Thickness Breach on 11-V-102
(Interactive Test Vessel)

1. Background & NDT Ultrasonic Survey Defect Discovery

During routine ultrasonic thickness gauging of the 1st Stage High-Pressure Separator (11-V-102), an acute wall thinning anomaly was detected at location **Shell Test Section** (Bottom Knuckle). Base material is **SA-516**. The field inspector recorded a measured wall thickness of **138.20 mm** against an accelerated corrosion rate of **0.75 mm/yr**.

2. Statutory Code Calculations (ASME Sec VIII Div 1 & API 510)

Verification using the ASME Sec VIII Div 1 UG-27 cylindrical shell formula yielded the following findings:

Parameter / Statutory Metric	Engineering Finding & Status
Design Pressure (P)	14.5 MPa (145.0 barg)
Minimum Required Thickness (t_min)	163.19 mm (ASME UG-27)
Actual Measured Thickness (t_actual)	138.20 mm at Point Shell Test Section
Safety Margin Delta (t_actual - t_min)	-24.99 mm [STATUTORY CODE BREACH]
API 510 Remaining Safe Service Life	-33.32 Years (BELOW ZERO)
Recommended Derated MAWP	123.7 barg (Immediate operational limit)

3. CVC Guidelines & Emergency Procurement Justification

CVC Circular No. 02/02/2004 & DOP Clause 4.2 Emergency Single-Source Procurement.

4. Recommended Action & Financial Sanction

Immediate emergency weld overlay repair and derated operation at 123.7 barg.

Estimated Total Expenditure: Rs. 88.0 Lakhs (Covered under Refinery Turnaround Capex Budget Code HCU-2026-CAP).

5. Submission & Approval Block

Prepared By:	Verified By:	Approved By:
Sr. Inspection Engineer Emp ID: 448102	Chief Manager (Integrity) Emp ID: 310294	Executive Director & Refinery Head Competent Financial Authority (DOP 2.1)